

The Great Pacific Garbage Patch

The Great Pacific Garbage Patch (also known as the Pacific Trash Vortex) is actually composed of two garbage patches: the Western Garbage Patch, located near Japan, and the Eastern Garbage Patch, located between Hawai'i and California. The reason all the trash accumulates is that it is not biodegradable; instead of breaking down, it just breaks into smaller pieces, killing aquatic life. Most of the garbage patch is made up of microplastics, which are really just tiny pieces of plastic and can be just as dangerous as macroplastics.

Of the 1.8 trillion pieces of trash accumulated in the Great Pacific Garbage Patch, it is split into two main sources: land-based and water-based. About 80% of the mass of the trash in the Great Pacific Garbage Patch is from land sources, and 20% is from the sea. Most of the trash is fishing supplies. Some of the reasons the plastics in the Great Pacific Garbage Patch are so dangerous are that turtles mistake plastic bags for jellyfish and get stuck, and albatross mistake plastic pellets for fish eggs and feed them to their young, killing the next generation of albatross. Other, larger marine animals get trapped in nets and die.

Many fish consume microplastics from the Great Pacific Garbage Patch and, in turn, pass them up the food chain, all the way to humans, resulting in their consumption of microplastics and long-term consequences. According to the provided sources, there is no great solution to the Great Pacific Garbage Patch other than slowly cleaning the trash and raising awareness so people stop dumping trash in the ocean.